UNIVERSITY OF SCIENCE
FACULTY OF INFORMATION TECHNOLOGY

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KCPM - 21KTPM2 - MID-TERM

COURSE: SOFTWARE TESTING

Requirement 8: Data Generation Report

18/08/2024 – 24/08/2024

COURSE CODE: CSC13003
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Ho Chi Minh City – 2024

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PROJECT INFORMATION

Course Code: CSC13003

Course Name: Software Testing

Title: CS423 – CSC13003 – Software Testing CASE STUDY (PROJECT)

Topic: Requirement 8: Data Generation Report

Description:

Software Under Testing: Test Management Website, currently in development with planned releases as follows:

- Prototype: Static web interfaces showcasing the main business functions.
- Build 1: Dynamic system retrieving information from a database, focusing on information display.
- Build 2: Complete system enabling users to manage the software testing process.

Requirement 8: Data Generation: Use data generation tools to create meaningful test data, including users, projects, test cases, and bugs.

Faculty Instructors: Ms. Trần Thị Bích Hạnh, Mr. Trần Duy Hoàng, Mr. Trương Phước Lộc, Mr. Hồ Tuấn Thanh

DATA GENERATION REPORT

I. Introduction:

This report outlines the process used to generate meaningful data for testing a software application. The data generated includes 30 users, 10 projects, 100 test cases per project, and 30 bugs per project.

II. Data Generation Tool:

1. *Tool Used:*

Mockaroo (https://www.mockaroo.com/)

2. *Tool Type:*

Free data generation tool with a user-friendly interface for creating customized datasets.

3. *Reason for Selection:*

Mockaroo was chosen because it allows for highly customizable field generation and supports a wide range of data types, making it ideal for generating the complex datasets required for this project.

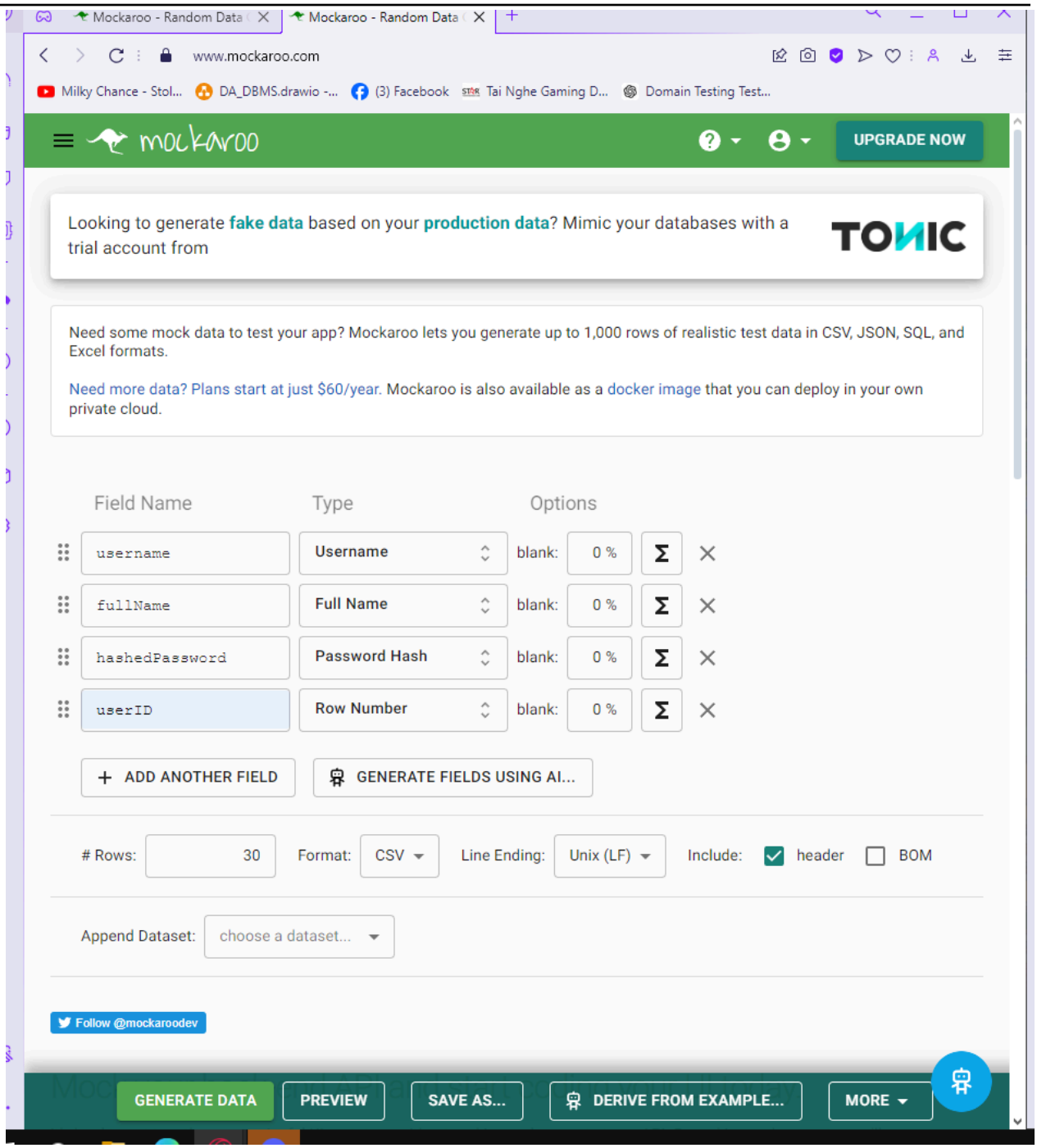
III. Data Requirements

1. *Users:*

Number of Records: 30

Fields Generated:

- `Username`: A unique username for each user.
- `Full Name`: A realistic full name.
- `hashedPassword`: A password that meets specific security requirements.
- `userId`: A reference to the user.



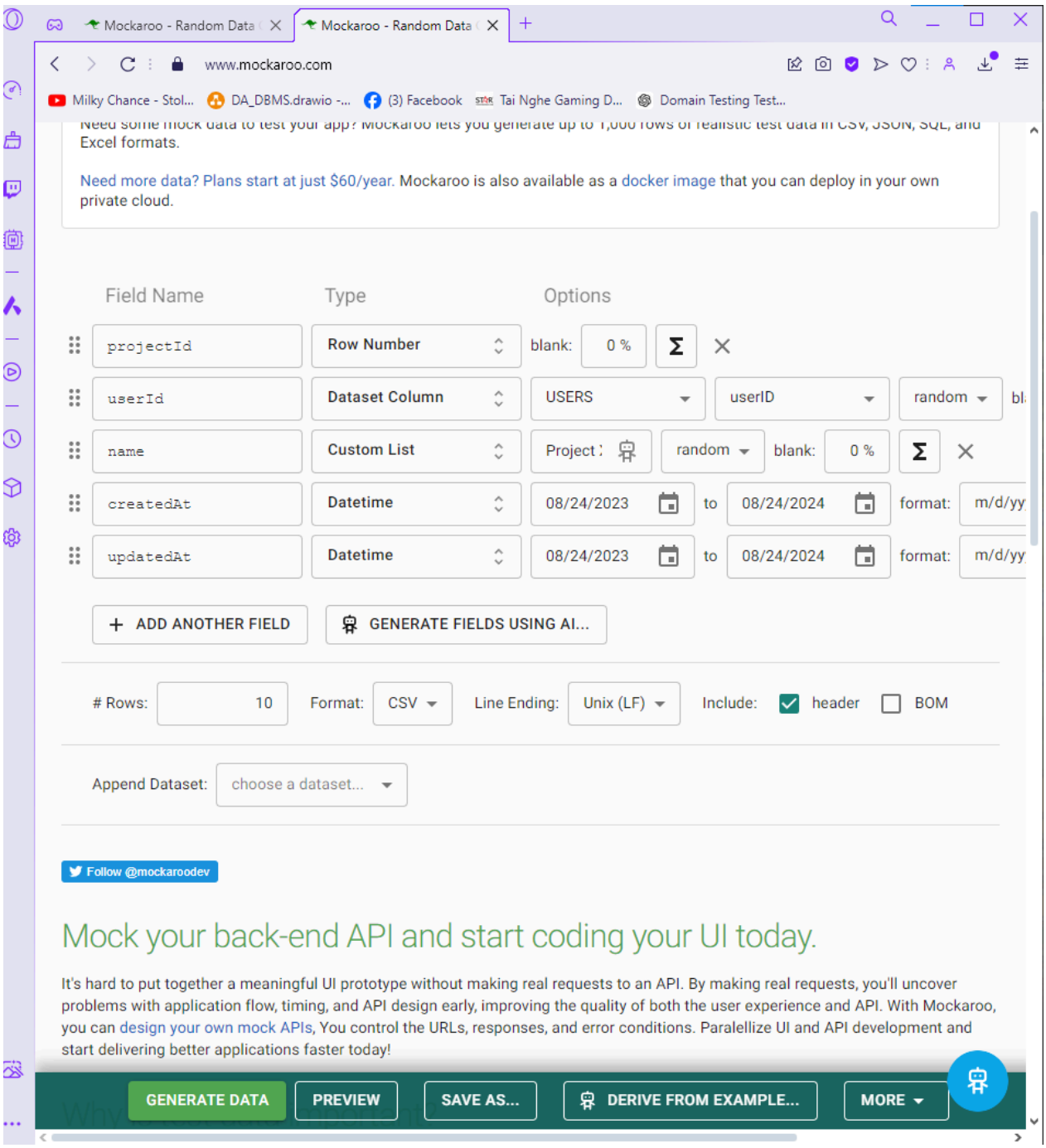
2. ***Projects:***

Number of Records: 10

Fields Generated:

- `projectId`: A unique identifier for each project.

- `userId`: A reference to the user responsible for the project.
- `name`: The name of the project, ensuring it is descriptive and relevant.

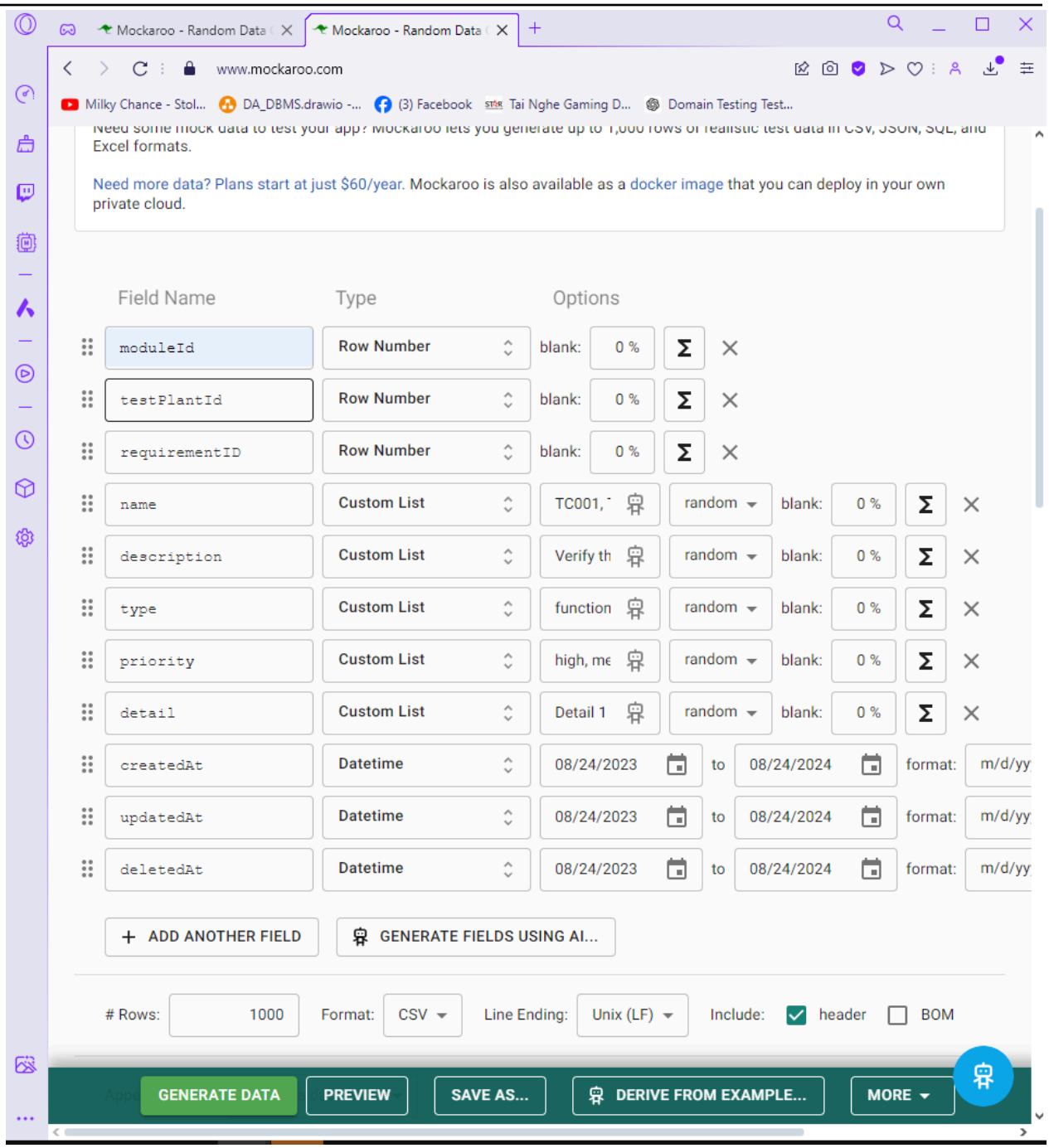


3. *Test Cases:*

Number of Records: 1000 (100 test cases per project)

Fields Generated:

- `projectId`: Configured to select the `id` from the previously created project dataset, ensuring that each test case is associated with an existing project.
- `moduleId`: An identifier for the module associated with the test case.
- `testPlanId`: An identifier for the test plan associated with the test case.
- `requirementId`: An identifier for the requirement associated with the test case.
- `name`: A descriptive name for the test case.
- `description`: A brief description of the test case.
- `type`: The type of test (e.g., functional, performance).
- `priority`: The priority level of the test case (e.g., high, medium, low).
- `detail`: Additional details relevant to the test case.
- `createdAt`: The date the test case was created.
- `updatedAt`: The date the test case was last updated.
- `deletedAt`: Indicates whether the test case was deleted.

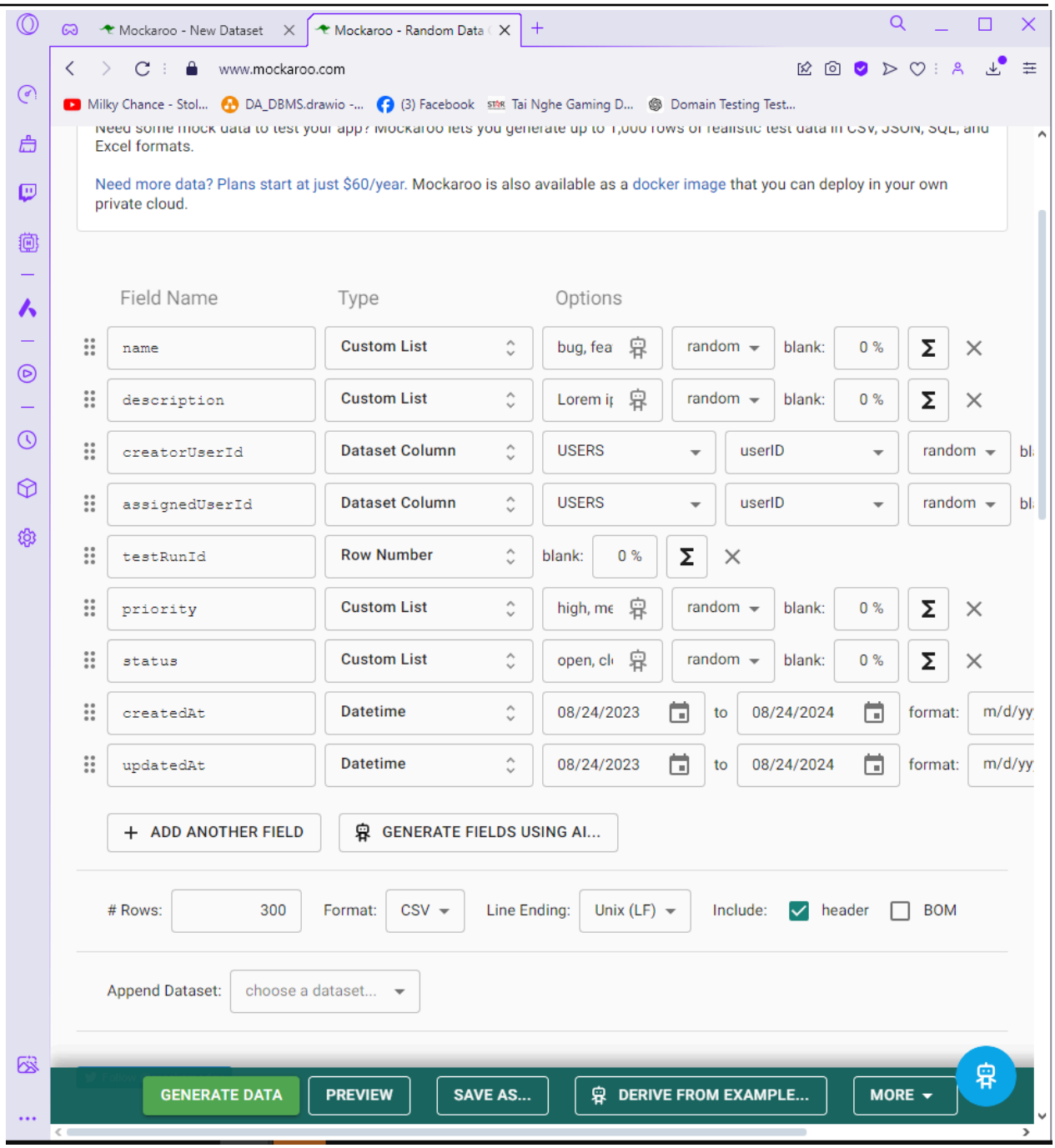


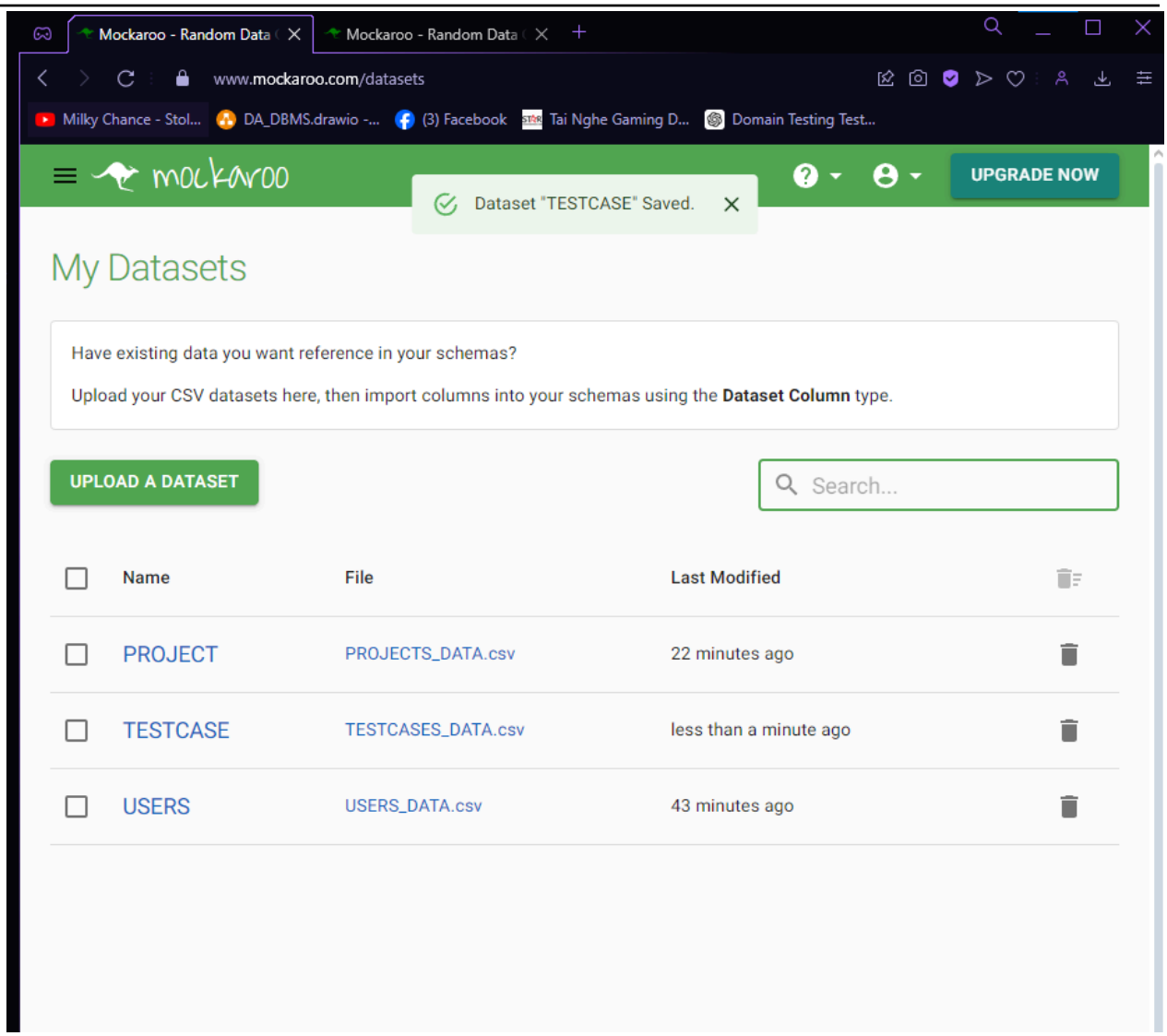
4. *Bugs*

Number of Records: 300 (30 bugs per project)

Fields Generated:

- `projectId`: Configured to select the `id` from the previously created project dataset, ensuring that each bug is associated with an existing project.
- `name`: A descriptive name for the bug.
- `description`: A brief description of the bug.
- `creatorUserId`: The ID of the user who reported the bug.
- `assignedUserId`: The ID of the user assigned to fix the bug.
- `testRunId`: The ID of the test run during which the bug was found.
- `priority`: The priority level of the bug (e.g., high, medium, low).
- `status`: The current status of the bug (e.g., open, closed).
- `createdAt`: The date the bug was reported.
- `updatedAt`: The date the bug was last updated.


Mockaroo - New Dataset
Mockaroo - Random Data
www.mockaroo.com
Milky Chance - Stol...
DA_DBMS.drawio -...
(3) Facebook
Tai Nghe Gaming D...
Domain Testing Test...
Need some mock data to test your app? Mockaroo lets you generate up to 1,000 rows of realistic test data in CSV, JSON, SQL, and Excel formats.
Need more data? Plans start at just $60/year. Mockaroo is also available as a docker image that you can deploy in your own private cloud.
Field Name
Type
Options
name
Custom List
bug, fea
random
blank:
0 %
description
Custom List
Lorem ip
random
blank:
0 %
creatorUserId
Dataset Column
USERS
userID
random
assignedUserId
Dataset Column
USERS
userID
random
testRunId
Row Number
blank:
0 %
priority
Custom List
high, me
random
blank:
0 %
status
Custom List
open, cl
random
blank:
0 %
createdAt
Datetime
08/24/2023
to
08/24/2024
format:
m/d/yy
updatedAt
Datetime
08/24/2023
to
08/24/2024
format:
m/d/yy
ADD ANOTHER FIELD
GENERATE FIELDS USING AI...
Rows:
300
Format:
CSV
Line Ending:
Unix (LF)
Include:
header
BOM
Append Dataset:
choose a dataset...
GENERATE DATA
PREVIEW
SAVE AS...
DERIVE FROM EXAMPLE...
MORE



We also need to save dataset to match with some values

REFERENCES

Supporting Tools and Software:

No	Function	Tool
[1]	System	Opera GX
[2]	Web	Opera GX
[3]	Project Management	Discord, Messenger, Google Drive

References:

[0]https://www.thedataschool.co.uk/elena-raluca-mazareanu/how-to-in-mockaroo-create-related-datasets-with-a-foreign-key/

[1] https://www.youtube.com/watch?v=S_oYFGhZSkQ